ISMAYIL YUSIFLI

Computer Science Student | AI/ML Engineering | Computer Vision
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PROFILE

Computer Science student at UFAZ focused on AI/ML engineering, applied computer vision, and production-oriented AI systems. Co-Founder of Guardian-X, contributing to the computer-vision pipeline behind AZira using PyTorch, FastViT, ONNX, and TensorFlow Lite. Hands-on experience spans deep learning, classical machine learning, semantic retrieval, natural-language inference, LLM-powered applications, model evaluation, backend APIs, automated testing, and AI deployment from experimentation through usable applications.

EDUCATION

French-Azerbaijani University (UFAZ) — BSc in Computer Science

2023 — 2027
Cumulative average: **14.485/20** | Official equivalent academic standing: **98/100**

EXPERIENCE

Co-Founder — Guardian-X | AI/ML Engineering

Aug 2025 — Present
Baku, Azerbaijan | Hybrid

- Contribute to the AI/ML engineering behind **AZira**, an AI-powered pest detection application publicly available on Google Play.
- Developed and evaluated a **FastViT-T12 image-classification pipeline** for pest recognition across **15 classes** using approximately **18,000 images**.
- Worked on dataset preparation, corrupted-image filtering, class mapping, transfer learning, model training, validation, testing, and generalization checks across train, validation, and test splits.
- Converted the trained PyTorch model through **ONNX and TensorFlow Lite** for mobile integration and investigated model-conversion issues during deployment preparation.
- Tested mobile conversion variants, identified unreliable float16 behaviour, and retained the working float32 model for more reliable integration.
- Contributed to AI integration planning, product iteration, pest knowledge-base development, and coordination between the model pipeline and the broader mobile product.

Machine Learning Intern — FlyRank AI

Jul 2026 — Present
Remote | Self-paced, project-based

- Built and executed an end-to-end machine-learning pipeline on approximately **30,000 records**, comparing a rule-based ranking approach with a Random Forest model.
- The Random Forest achieved **Precision@50 of 0.74 versus 0.24 for the rule-based baseline**.
- Investigated feature selection, decision-tree depth, feature importance, target leakage, model interpretability, generalization, and validation methodology.
- Maintained executable notebooks and reproducible experiments, validating changes by rerunning pipelines and checking outputs.

C/C++ Developer Intern — Intern Intelligence

Feb 2025 — Mar 2025
Remote

- Built an advanced mathematical-functions library using C and C++.
- Developed a multithreaded task scheduler for managing and executing concurrent tasks.
- Worked with data structures, algorithms, debugging, multithreading, concurrency, CMake, and testing.

TECHNICAL SKILLS

Machine Learning & AI: PyTorch, scikit-learn, NumPy, Pandas, OpenCV, CNNs, FastViT, Sentence Transformers, semantic retrieval, multilingual embeddings, natural-language inference, image classification, transfer learning, model evaluation, experiment design
Programming: Python, TypeScript, JavaScript, C++, C, Java, SQL
Model Deployment & Engineering: FastAPI, Flask, REST APIs, ONNX, TensorFlow Lite, Docker, Git, Modal, Vercel, Render, debugging, automated testing
AI Engineering Tools: Gemini API, Google GenAI SDK, Codex, Claude Code, Gemini, DeepSeek; structured LLM outputs, prompt design, agentic workflows, AI-assisted debugging, testing, architecture work, and code review

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SELECTED PROJECTS

LexProof — Evidence-First Legal Claim Verification | LexHack 2026

Python, FastAPI, PyMuPDF, Sentence Transformers, Multilingual E5, NLI, Next.js, TypeScript, Tailwind CSS, Modal, Vercel

- Independently designed and built an end-to-end legal AI verification platform that checks legal claims against **authoritative EU GDPR text** and returns evidence-backed **SUPPORTED, CONTRADICTED, or INSUFFICIENT EVIDENCE** verdicts.
- Built a Python/FastAPI legal-document processing pipeline using **PyMuPDF**, correctly identifying all **99 GDPR articles** while preserving article, chapter, section, page, and source metadata.
- Implemented multilingual semantic retrieval using **`intfloat/multilingual-e5-small`**, including passage and query embeddings, ranking, article-level deduplication, and a persistent cached retrieval index.
- Integrated **`MoritzLaurer/multilingual-MiniLMv2-L6-mnli-xnli`** for multilingual natural-language inference and combined it with deterministic verification logic for concrete conflicts such as statutory deadline mismatches.
- Designed three-way verification to distinguish **actual contradiction from absence of evidence**, preventing unsupported legal claims from being incorrectly classified as false.
- Diagnosed and fixed an NLI **context-dilution problem** by evaluating complete numbered legal paragraphs alongside larger retrieved passages while preserving original article metadata.
- Built a responsive **Next.js/TypeScript/Tailwind** frontend showing verdicts, confidence information, relevant articles, quoted primary-law evidence, explanations, and official EUR-Lex source links.
- Created automated regression and integration coverage with **36 backend tests, 8 verification benchmark cases, and 22 frontend/browser/API-contract tests**.
- Deployed the frontend on **Vercel** and the FastAPI/ML inference pipeline on **Modal**, including model packaging, retrieval-index persistence, cold-start handling, and production debugging.
- Delivered the project independently from concept through architecture, implementation, testing, deployment, documentation, demo recording, and final hackathon submission.

From-Scratch Softmax Regression & Neural Network Classifier

Python, NumPy, Matplotlib

- Built softmax regression and a one-hidden-layer neural network from scratch with NumPy, implementing forward propagation, loss computation, backpropagation, and training logic without relying on high-level deep-learning frameworks.
- Implemented mini-batch training, validation checkpointing, reproducible experiment runners, and optimization using SGD, Momentum, and Adam.
- Ran repeated-seed and hidden-layer-capacity experiments across Gaussian, moons, and handwritten-digit datasets to compare linear and nonlinear models using accuracy and cross-entropy loss.

Azerbaijani Handwritten Character Recognition

Python, CNN, OpenCV, Flask

- Built an end-to-end computer-vision system for recognizing handwritten Azerbaijani characters using a custom collected dataset, covering dataset preparation, preprocessing, model training, evaluation, and inference.

- Developed an OpenCV-based preprocessing workflow and trained a CNN classifier, evaluating predictions using classification metrics.
- Integrated the trained model into a Flask-based interactive prediction demo, turning the model into a usable web application.

CourseFlow — AI Coursework Execution Planner

Gemini, Google GenAI SDK, TypeScript, Node.js, Express

- Built an AI-powered coursework execution planner that converts uploaded coursework into **evidence-backed requirements, decomposed tasks, workload estimates, and a realistic execution schedule**.
- Implemented page-aware document extraction with structured Gemini outputs that preserve **filename, page, and source-snippet evidence** for extracted requirements and deadlines.
- Combined LLM-based extraction and task decomposition with **deterministic dependency scheduling**, workload-feasibility checks, mandatory and optional task handling, and shortfall detection.
- Developed **constraint-aware replanning** that preserves unaffected scheduled work while rescheduling impacted tasks and propagating unfinished dependencies.
- Added Gemini fallback behaviour, API integration, deployment, and debugging across local and hosted environments.
- Diagnosed extraction and evidence-traceability issues and improved the pipeline so generated requirements remained connected to their original source material.

AI & ENGINEERING TRAINING

- **AI Engineering Program** — AI Academy, Ongoing
- **Math4AI Track** — AI Academy, Completed
- **Claude with the Anthropic API** — Anthropic Academy
- **AI Fluency: Framework & Foundations** — Anthropic Academy
- **Certificate of Proficiency — Java & Back-End Web Development** — Code Star
- **Mendix Rapid Developer Certification** — Mendix

ACHIEVEMENTS

- **Winner — "Əlaçı" Scholarship Program, PAŞA Holding — 2026**
- **3rd Place** — Siemens Sustainable Design Hackathon, Guardian-X Team
- **Honorable Mention** — 2025 ICPC Azerbaijan Qualification
- **PASHA Hackathon 6.0** — Built a retail waste-management solution with a team

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LANGUAGES

Azerbaijani: Native • **English:** Advanced / C1 • **Turkish:** Intermediate • **French:** Elementary / A2